

Dears, some of you asked me about backend resources, here are two books and a YouTube playlist:

- Physical design essentials by kosrow glosan
- ADVANCED ASIC CHIP SYNTHESIS Using Synopsys® Design Compiler" and PrimeTime®
- <https://youtube.com/playlist?list=PL1h5a0eaDD3pimcMlzW15RpW02HPzIziL&si=zLqtqn7MqBZptF3i>

Extra Designs to Practice & **Verify**:

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- 1- https://hdlbits.01xz.net/wiki/Problem_sets#Verilog_Language
 - 2- <https://www.fpga4student.com/p/verilog-project.html>

3- AMBA® APB

Protocol: https://www.eecs.umich.edu/courses/eecs373/readings/IHI0024C_amba_apb_protocol_spec.pdf or <https://documentation-service.arm.com/static/60d5b505677cf7536a55c245?token=>

>> Here is a post visualizing the APB protocol shared by our colleague Mohamed Samir:

https://www.linkedin.com/posts/amradelm_asic-pnr-sta-activity-7235662304694992896-r7VD?utm_source=share&utm_medium=member_desktop

4-

AMBA® AHB Protocol: <https://developer.arm.com/documentation/ih0033/latest>

5-

AMBA® AXI4-Lite

Interface: <https://www.realdigital.org/doc/a9fee931f7a172423e1ba73f66ca4081> or <https://developer.arm.com/documentation/ih0022/latest/>

5- UART: <https://nandland.com/uart-serial-port-module/>

6- AMD 7-Series and Zynq 7000 SoC blocks: <https://docs.xilinx.com/r/en-US/ug953-vivado-7series-libraries/Introduction>

-- The following are large/major blocks that will take effort to design:

- 1- DSP48E1
- 2- FIFO18E1
- 3- FIFO36E1

4- RAMB18E1

5- RAMB36E1

Note: After each block description, you will find an external link that you can click on to learn more about the specifications of the block

7- Microsemi FPGA blocks: https://www.microsemi.com/document-portal/doc_view/135041-rtg4-macro-library-user-guide

-- The following are large/major blocks that will take effort to design:

1- RAM1K18_RT

2- RAM64x18_RT

3- MACC

Big projects that you can work on as a team (can be a graduation project):

1- MIPI DIS: <https://picture.iczhiku.com/resource/eetop/WHISUKJqgiqojbxm.pdf>

2- Bluetooth low energy: <https://www.bluetooth.com/specifications/>

3- CPHY, DPHY

4- PCIE Controller

5- UCIE

6- CXL

7- RISC-V Processor

Note that the above projects are big projects so you do not have to fully implement all of its features. Instead, start small and develop more as time permits.

SV DPI recording by my assistant Mustafa Ibrahim:

Slides walkthrough: <https://drive.google.com/file/d/1rrQWGK4YpkA-3nP4LZPn1H4ttBqedvze/view?usp=sharing>

Demo: <https://drive.google.com/file/d/11s2HPtk5akXCSglGsiV71iF5KYSJN4S-/view?usp=sharing>